
25-g Accelerometer

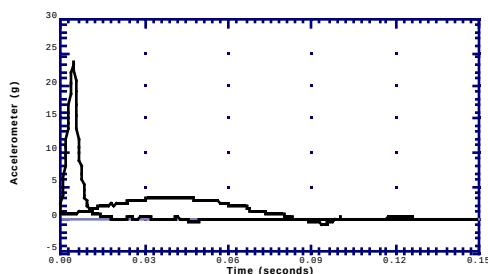
(Order Code ACC-BTA or ACC-DIN)

The 25-g Accelerometer can be used for a wide variety of experiments and demonstrations, both inside the lab and outside. Some suggested activities in the laboratory are:

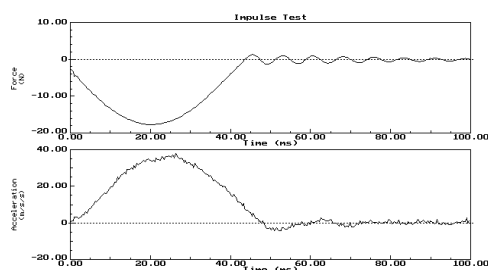
- Measure acceleration during a collision.
- Measure acceleration as you swing the accelerometer in a horizontal circle. Study the relationship between acceleration and period, acceleration and radius, and acceleration and mass.
- Measure acceleration as you swing the Accelerometer in a vertical circle.
- Put the Accelerometer under your belt buckle, and jump up and down. Measure the acceleration as you land, both with your knees flexed and with your knees held more stiff.
- Have a packing contest. Challenge people to pack the Accelerometer in a box and to minimize the deceleration when it is dropped from a specified height. Note that you need to orient the Accelerometer (and the box) so that the Accelerometer measures properly. It can only measure acceleration along the line marked by the arrow.

If you have a lab interface that allows you to collect data away from the lab, you might try to measure acceleration

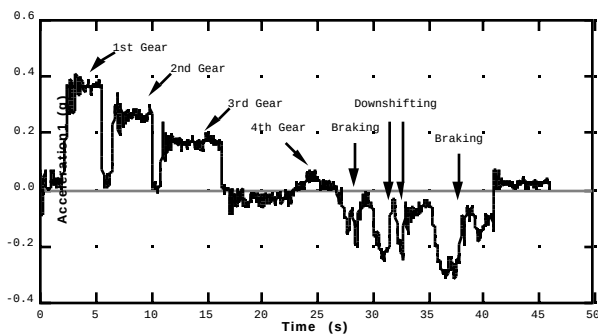
- in a car;
- on amusement park rides;
- on playground apparatus, such as merry-go-rounds;
- of bicyclists, skiers, or bungee jumpers, etc.



Collision of a dynamics cart with and without a spring bumper



Force and acceleration measured at the same time during a collision



Using the Accelerometer in a manual-shift car

How the Accelerometer Works

The Accelerometer senses acceleration using an integrated circuit (IC) originally designed to control the release of air bags in an automobile. This IC is micro-machined with very thin “fingers” carved in silicon. These fingers flex when accelerated. They are arranged and connected like the plates of a capacitor. As the fingers flex, the capacitance changes, and a circuit included in the IC monitors the capacitance, converting it into a voltage. The op-amp circuit on our circuit board amplifies and filters this signal. This output can be read by any of the following lab interfaces available from Vernier:

- Vernier LabPro™ (for use with computers or TI Graphing Calculators)
- Texas Instruments CBL™ 2 or CBL System
- Universal Lab Interface (ULI)
- Serial Box Interface
- MultiPurpose Lab Interface (MPLI)

The Accelerometer measures acceleration along the line marked by the arrow on the circuit board. Accelerations are normally measured in either meters per second per second (m/s/s) or g’s. One g is the acceleration due to gravity at the earth’s surface, or 9.8 m/s/s. Our Accelerometer will measure accelerations in the range of -25 g (-250 m/s/s) to +25 g (250 m/s/s). Even though this is a fairly large acceleration, it is easy to produce accelerations larger than this in collisions. In fact, dropping the Accelerometer on a hard surface from even a few centimeters can produce accelerations of a hundred g’s. The Accelerometer will not be damaged by accelerations up to 500 g’s.

The Accelerometer is based on the ADXL50 from Analog Devices. There is inherent noise in the ADXL50. This limits the use in low acceleration environments. The noise is typically on the order of 2.5 m/s/s peak to peak. Therefore, best results are achieved if the experiments are designed around magnitudes or changes exceeding 9.8 m/s/s. The offset voltage (voltage output at 0 m/s/s) will drift with temperature. Performing a calibration prior to the experiment will provide the best results.

Vernier produces a different accelerometer, the Low-g Accelerometer (order code LGA-BTA or LGA-DIN), which is better for experiments with small accelerations. It has a range of ± 50 m/s/s (± 5 g’s). Its improved sensitivity and reduced noise make it a better choice for experiments with small accelerations.

NOTE: This product is to be used for educational purposes only. It is not appropriate for industrial, medical, research, or commercial applications.

If you purchased the ACC-BTA, you will find a DIN to BTA adapter in the box with your sensor. Use the adapter to connect the sensor to a LabPro, CBL 2 or CBL.

Calibration

Calibration may be done using the acceleration due to gravity. To calibrate the sensor for measuring acceleration in the horizontal direction, position the Accelerometer with the arrows pointing down for the first calibration point. Define this as -9.8 m/s/s or -1 g. Rotate the Accelerometer so the arrows point up and use the reading for the second calibration point. Define this as +9.8 m/s/s or +1 g. The Accelerometer will then read 0 with no acceleration when held horizontally.

If you want to calibrate for measuring acceleration in the vertical direction, follow the procedure above, but define the first calibration point as 0 g or 0 m/s/s and the second point as 2 g or 19.6 m/s/s.

Using Multiple Accelerometers

You may want to simultaneously measure the signals from two or three Accelerometers mounted at right angles. Two mounted at right angles allow you to study acceleration in a plane. This works well, for example, in an automobile. You can mount three at right angles to study acceleration in every direction. Calibrate each Accelerometer for measuring horizontal acceleration. Mount them at right angles. Create a "New Column" using your lab interface program to calculate the square root of the sum of the squares of the accelerations. This value will be equal to 1 g when the combination accelerometer has no acceleration, and zero when it is in free fall. The orientation of the combination accelerometer does not matter.

Specifications	Power: 10 mA @ 5 VDC	Range: ± 245 m/s/s (± 25 g)
	Accuracy: ± 2.45 m/s/s (± 0.25 g)	Frequency Response: 0 - 100 Hz

Using the Accelerometer with a Computer

This sensor can be used with a Macintosh or PC computer and any of the following lab interfaces: LabPro, Universal Lab Interface, or Serial Box Interface. Follow these general procedures to use the Accelerometer with a computer:

1. Connect the Accelerometer to the appropriate port on the interface.
2. Start the data collection software on the computer. If you are using a Power Macintosh or Windows computer, run the *Logger Pro*™ software. If you are using an older Macintosh, DOS, or Windows 3.1 computer, run the Data Logger program.
3. Open an experiment file in the *Logger Pro* or Data Logger folder, and you are ready to collect data.

Using the Accelerometer with TI Graphing Calculators

This sensor can be used with a TI Graphing Calculator and any of the following lab interfaces: LabPro, CBL 2, or CBL. Follow these general procedures to use the Accelerometer with a graphing calculator:

1. Load a data-collection program onto your calculator:
 - LabPro or CBL 2: Use the DataMate program. This program can be transferred directly from LabPro or CBL 2 to the TI Graphing Calculator. Use the calculator-to-calculator link cable to connect the two devices. Put the calculator into the Receive mode, and then press the Transfer button on the interface.
 - Original CBL: Vernier has several programs that support this sensor and other sensors. When using the Accelerometer with the original CBL, we recommend the PHYSICS program. This program is available free on our web site at www.vernier.com. Our programs can also be obtained on disk. (Contact us for more information.) Load the program into a calculator using TI-GRAPH LINK.
2. Use the calculator-to-calculator link cable to connect the interface to the TI Graphing Calculator using the I/O ports located on each unit. Be sure to push both plugs in firmly.
3. Connect the Accelerometer to any of the analog ports on the interface. In most cases, channel CH 1 is used.
4. Start the data-collection program and you are ready to collect data.

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